Evolving Native-Noise Controllers for Safe Completion with Frozen Manipulation Policies

Implemented method, completed development evidence and a frozen Jar comparison

Sachin Mohan
University of the Witwatersrand
Johannesburg, South Africa
2699183@students.wits.ac.za

ABSTRACT

Can a compact controller recover safer task completion from a frozen generative manipulation policy under a small interaction budget? We implement an observation- and instruction-conditioned adapter that adds a bounded four-dimensional offset to native sampling noise. Its 176 parameters use measured robot state, executed action history and frozen image/text features; the family-specific VLA and native execution contract remain fixed. A bounded 320-call decoder audit on ManiGuard Jar and Clutter reproduces native sampling on all 32 same-noise identity pairs and confirms influence on the executed action prefix in all four tested directions per family. Median perturbed-prefix arm differences are .00319 and .00187 rad, respectively. Four full development episodes establish integration and timing, without a matched estimate of physical benefit. We freeze a 60-valid-episode allocation: four development episodes, eight shared initial episodes, sixteen descendant episodes each for evolution and random search, and sixteen independent checks of native, fixed, evolved and random-search controllers. All eight shared initial episodes end unfinished; the highest-utility constant never grasps. Search is running and independent comparison results are pending. A separate 48-call diagnostic finds gripper-coordinate influence at a stronger noise offset, with arm coupling and no physical benefit tested. Native-noise steering and evolutionary VLA steering have precedents. The contribution sought is an explained safe-completion benefit per total interaction, with competent learning comparisons, replication and transfer still required.

CCS CONCEPTS

• **Computing methodologies**;

KEYWORDS

evolutionary robotics, frozen policies, latent steering, safe manipulation

1 INTRODUCTION

Completing a manipulation task and respecting its physical specification are different outcomes. A robot may transport a jar with its lid open, or avoid every violation while never finishing. ManiGuard makes these distinctions measurable through task checks and passive runtime monitors [4]. We investigate whether a small controller can improve safe requested completion around its family-specific frozen vision-language-action policies.

The adapter steers the generator's native noise before decoding an action chunk, conditioning on current observations, the actual instruction and causal executed action history. Evolution optimizes a small parameter matrix from complete physical simulator outcomes. The VLA, feature encoder and native execution contract remain fixed. There is no learned safety filter, world model or online action veto.

Earlier restraint and feedback pilots [2] motivate this change without validating it. The earlier full-strength finalist achieved 4/8 safe completions with two violations, against identity's 5/8 with none. An adaptive query trigger never refreshed early. A saved-trace diagnostic verified a direction-preserving cap but found few trigger opportunities. These are historical results from different controllers and a legacy rendering profile, not controls for the present experiment.

We separate four questions: does adaptation help native execution; does conditioning help a fixed bias; does evolutionary retention and mutation help relative to other adaptation methods; and do fitted weights transfer? Our completed evidence establishes integration and decoder influence. A matched Jar comparison is in progress under a protocol frozen before comparative feedback. Its small sample can provide effect-size leads, not establish optimizer superiority or cross-family transfer. Shared initialization now supplies eight valid full episodes without safe completion on its two cases.

2 RELATED WORK AND CONTRIBUTION BOUNDARY

DSRL learns in the latent-noise space of a frozen diffusion policy, including generalist-policy adaptation [8]. Flow Reversal Steering maps coarse reference actions through a reversed flow and learns auxiliary noise policies; it also studies repeated or averaged noise across chunk positions [7]. Principal Steering Subspaces constructs a low-dimensional basis from decoder responses and controls its leading directions with SAC [3]. Noise-space adaptation, chunk repetition and low-dimensional steering are therefore not novel by themselves. Our present basis is random and fixed, not response-informed.

Evolutionary VLA steering also has precedents. VLA-Pilot evolves action proposals under an inferred steering objective during inference [1]. Online-ES uses executed trajectory perturbations and feedback-weighted training to update model parameters [9]. Here the base VLA stays frozen while selection updates a reusable conditional adapter using specification-labelled

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episode outcomes. This defines an empirical question, not a priority claim. A conventional EA or optimizer substitution alone is insufficient novelty.

The intended contribution is a reproducible, mechanistically explained improvement in safe requested completion per total interaction. Evidence must distinguish useful steering from a tuned constant, and evolutionary advantage from random search and competent ordinary learning. One fitted adapter working with several family checkpoints is the longer-term target. Reusing its interface with per-family refitting is a weaker claim than transferring the same weights.

3 PHYSICAL EVALUATION BOUNDARY

At query time τ, the frozen family policy decodes native Gaussian noise $\epsilon_\tau \in \mathbb{R}^{16\times 32}$ from two RGB views, robot input and the prompt. Native output transforms produce physical joint-target/gripper commands. The first eight are executed unless native termination occurs sooner. The adapter changes noise rather than action horizon.

Let V be raw native `ltl_violated`, without engagement gating. Requested completion C_{req} requires native success and the pinned family checklist. For Jar, closure must be observed strictly before lifting, remain true at every off-support observation and at the end, and lifting must occur. The final native held/intersection goal must also hold. This stricter sequence check is reported beside the unmodified benchmark verdict. Safe completion and training utility are

$$S_{\text{req}} = C_{\text{req}} \wedge \neg V, \qquad U = \mathbf{1}(S_{\text{req}}) - 2\mathbf{1}(V). \tag{1}$$

A violating completion scores -2; unfinished nonviolating execution scores zero. This is a declared penalized objective. Reporting retains all four joint completion/safety categories, raw violations and denominators. Contact/grasp indicators contextualize unfinished episodes; fewer violations can reflect inactivity.

The monitor supplies evaluation labels and never changes commands. Monitor predicates, object poses, task/family/checkpoint identifiers and goal state are excluded from the adapter. Timeouts are valid outcomes, not infrastructure failures, and do not alone identify their physical cause.

4 NATIVE-NOISE CONTROLLER

4.1 Causal features

The feature vector $\phi_\tau \in [-1,1]^{44}$ comprises a bias, seven arm positions divided by π, two measured finger positions divided by .04 m, seven previous executed arm targets minus current arm positions divided by .2 rad, and three end-effector displacements since the previous query divided by .05 m. Robot/history coordinates are clipped. Initially the previous target equals measured joints and displacement is zero.

The remaining 24 coordinates are eight each from overview RGB, wrist RGB and the actual instruction. Frozen CLIP ViT-B/32 [6] produces 512-dimensional embeddings. Each embedding is normalized to unit Euclidean norm. Fixed Gaussian-QR projections, seeded 20261017, produce eight coordinates, multiplied by $\sqrt{512}$, clipped to $[-5,5]$ and divided by five. No feature fitting uses study outcomes. Images are encoded at every native query and text once per episode, including in the native reference. The conditioning path is implemented; its physical value is unestablished.

4.2 Bounded steering with native stochasticity

For genome $W \in [-1,1]^{4\times 44}$, define $c_\tau = \tanh(W\phi_\tau)$ and

$$z_{\tau,h} = \epsilon_{\tau,h} + .35Bc_\tau, \qquad h = 1,\dots,16. \tag{2}$$

The first seven rows of $B \in \mathbb{R}^{32\times 4}$ are a reduced QR basis of a Gaussian 7×4 matrix with PCG64 seed 20261019. Remaining rows are zero. The offset repeats across the chunk and has Euclidean norm at most $.35\sqrt{4} = .70$ per position in noise coordinates. This bounds sampling input, not joint motion or violation probability. Gripper/padding noise are not directly steered, but decoder coupling can affect their outputs. No persistence, learned noise scale or activation gate is used.

For $W = 0$, an explicit branch calls unchanged native `Policy.infer` without an explicit-noise argument. Nonzero controllers preserve the native RNG split and Gaussian draw before adding the offset. A zero noise tensor would define a different policy. Receipts retain RNG transitions, noise/observation hashes, offsets, decoded physical chunks and source hashes. The runtime logs steered proposals and final environment inputs. It does not buy an additional unsteered chunk at every query; same-noise comparisons are confined to the bounded decoder audit.

4.3 Matched search

Shared initialization evaluates identity, positive and negative constants, and a uniform random conditional matrix on two cases. Constants set only $W_{0,0} = \pm\operatorname{atanh}(.5)$, yielding $\pm.175$ along the first basis direction. The random matrix uses seed 20261019. Both methods receive these same eight episodes, counted once globally and attributed to both information budgets. The better constant becomes the fixed reference, with ties selecting positive. This tunes two declared constants, not an extensive fixed-policy baseline.

Each method evaluates two generations of four candidates on two common cases per generation. Both freshly re-evaluate the previous cohort's best candidate. EA adds three children from alternating top-two parents using iid Gaussian weight mutation with standard deviation .25 and clipping to $[-1,1]$. Random search instead draws three uniform matrices from the same domain, with the same retained-incumbent cost. Ranking uses summed utility and stable identifiers for ties. Only the last complete cohort supplies each finalist, preventing best-of-repeat promotion. Both method RNGs use seed 20261020. Descendant execution interleaves methods at each candidate slot, using EA/random at even slots and random/EA at odd slots. Lineage and pending/completed evaluations are persisted.

5 PROSPECTIVELY FROZEN PHYSICAL PROTOCOL

5.1 Scope and native execution

Jar and Clutter provide development exposure; Cabinet and Dusty remain physically untouched by this study. Jar/Stack/Lid/Clutter have programme-level history. Jar is the primary pilot and Clutter the declared capacity fallback. The selected protocol was frozen at

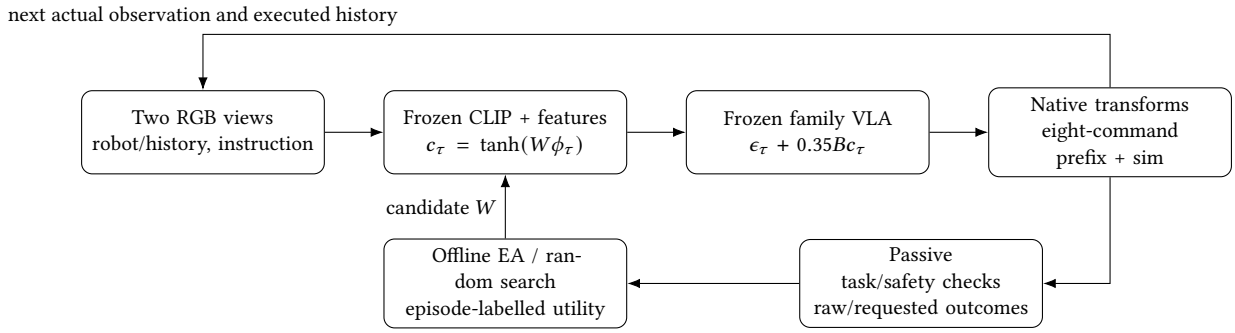


Figure 1: Implemented information boundary. Candidate weights are fixed within an episode; the native Gaussian draw remains present. Outcome labels guide selection of later candidates without vetoing execution. All conditions pay the same feature extraction cost. The VLA also receives its unchanged native observation inputs. No learned safety filter or world model is deployed.

Algorithm 1 Frozen comparison after four development episodes

Require: Pinned policy, features, domain, cases, seeds and budget
Evaluate four shared controllers on two cases (8 episodes)
Select fixed reference from the two constants
Initialize EA and random search with the same scored data
for generation $g = 1, 2$ **do**
    Propose both four-candidate cohorts, including incumbents
    for candidate slot $j = 0, 1, 2, 3$ **do**
        for method in the frozen alternating slot order **do**
            Evaluate its candidate on two common cases
            Verify both physical receipts and record scores
        end for
    end for
    Rank each complete cohort for subsequent selection
    Preserve all attempts, ancestry, RNG and physical records
end for
Seal native, fixed and the two final-cohort finalists
Check four conditions on four blocks (16 episodes)
Report outcomes, block differences, activity and costs

04:27:23 UTC on 20 October before comparative feedback, using development timing and the authority criterion. It selects Jar. This pilot does not train or test one shared cross-family adapter.

Execution uses native 640 × 640 DLAA rendering, two actual camera views, the verified extended-finger Franka, assisted grasping, 20 Hz policy control, 60 Hz rendering and 120 Hz physics. Hard horizons are 2,000 controls for Jar and 1,100 for Clutter. Preprocessing, gripper binarization, native clipping and termination are unchanged. Jar uses checkpoint step 7400, revision `1d84eda07031`; Clutter uses step 7100, `f572e33bdb49`. Full revisions and file hashes remain in study records.

Jar training tasks are 0019/0009, with base seeds $6101910 + 10g + i$, generation $g = 0, 1, 2$ and task index $i = 0, 1$. Checking tasks 0016/0012/0011/0003 use seeds $6101940 + i$. These blocks are reserved from this optimizer's feedback. Historical validation allocation does not establish absence of earlier programme exposure; we conservatively describe them as exposed tasks with independent checking. The original main-study test partition remains excluded. Native per-scene sampling-seed derivation is retained.

Table 1: Frozen full-episode allocation. Shared initialization is acquired once. The sixteen-episode independent check is protected from adaptation and tuning.

Stage	Valid episodes
Development, Jar + Clutter	4
Shared initialization: four controllers, two cases	8
EA: two generations, four candidates, two cases	16
Random search: same structure and domain	16
Independent check: four conditions, four blocks	16
Planned total	60
Maximum starts with infrastructure replacements	64

5.2 Allocation and hardware balancing

Table 1 specifies the 60 planned valid full episodes. At most four genuine infrastructure failures may be replaced, yielding a 64-full-attempt ceiling. Every partial/invalid attempt and discarded candidate is charged. Valid timeouts and adverse scores cannot be replaced. Separate engineering used six starts: three failed during preparation and three reached 32 controls each. The 96-control engineering allowance is exhausted. Physics before callback attachment is unknown, not zero. The original 320-call decoder allowance is exhausted. A separately authorized 48-call follow-up, reported below, changes neither this physical allocation nor the controller used in the frozen comparison.

Two workers each colocate simulation and policy on one RTX 3090. Training case i uses worker $(i + g) \bmod 2$ for both methods. Denoting native, fixed, EA and random by N, F, E, R, checking orders are (N, F, E, R), (F, R, N, E), (E, N, R, F) and (R, E, F, N). Adjacent pairs run concurrently; positions zero and two use worker zero. Every condition therefore runs twice per GPU. One admission ledger enforces the global attempt ceiling. Process identities

and terminal receipts permit reattachment without duplicate physical launches.

Capacity protects 56 post-development episodes plus four replacement slots as 30 two-worker waves. Its per-wave bound is the maximum of 180 s, 1.25 times the slowest complete development process, and a startup-plus-native-horizon throughput estimate with the same margin. Including 900 s for lifecycle/preservation, Jar requires 15.008 h. The original window was insufficient; the user subsequently authorized the full comparison on this instance. The fixed cutoff is 19:30 UTC and independent local stop 20:00 UTC on 20 October. Horizons/rendering were not reduced to fit.

5.3 Analysis and uncertainty

The final comparison will retain all four requested completion and safety categories and raw benchmark completion per condition, with paired differences by checking block. Initial robot/image hashes are checked: common seeds do not prove identical camera observations. Contact/grasp indicators contextualize noncompletion. Cost curves use common cumulative physical control counts and retain all adaptation, failures and discarded candidates. Model calls, encoder work, elapsed time and allocated GPU-hours are separate counters. Shared episodes count once physically and once within each method's information budget.

Setup, decoder calls, idle allocation and preservation are charged. GPU-hour and running-cost estimates are upper bounds over elapsed wall time from the original provider start, counting both GPUs. The own-instance API does not expose verified billed runtime; a container restart is not credited as unbilled time. Four checking episodes per condition and one optimizer seed provide descriptive effect sizes, not a superiority or safety guarantee. Zero violations in n independent episodes implies an approximate 95% upper rate bound of $3/n$, not zero; four episodes are particularly weak evidence.

6 COMPLETED DEVELOPMENT EVIDENCE

6.1 Identity and decoder influence

Eight saved observations per family, from two development tasks, each use two native noise draws. Each draw receives two same-noise baselines and positive/negative offsets along four directions: ten calls and 320 total. All sixteen identity pairs per family pass the declared 10^{-5} absolute/relative tolerance. Full remote recovery also reproduces noise hashes and RNG transitions without new model calls.

Perturbed-prefix arm RMS ranges from .001131 to .006422 rad for Jar, median .003193, and .000666 to .006975 rad for Clutter, median .001868 (Figure 2). Native-resampling medians are .011827/.005118 rad. All four directions per family pass the frozen criterion: observation-level mean RMS at least .001 rad on at least half the observations, with two passing directions required. The smallest individual Clutter response need not exceed this observation-level threshold. No executed-prefix gripper flips occur among 128 directional probes per family. Coupling elsewhere is not excluded.

A 32-control Clutter integration check uses an untrained dense conditional matrix, exercising all 176 weights including 96 semantic weights. Removing semantic features offline changes its code by RMS .3781 on factual observations. This verifies a software path, not semantic competence or a physical counterfactual.

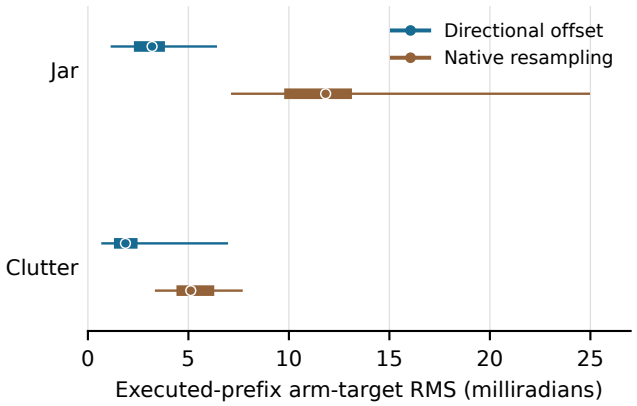


Figure 2: Arm-target RMS differences over the executed eight-command prefix. Thick intervals show quartiles, thin intervals the range and circles the median. There are 128 directional comparisons and eight native-resampling comparisons per family, clustered within eight saved observations. These are decoder responses, not independent physical trials or safety measurements.

Table 2: Full development episodes. C is requested completion, V raw violation, A controls and Q policy calls. Each row uses a different task/seed.

Family/controller	Task	A	Q	C	V	U
Jar native	0013	666	84	0	1	−2
Jar positive bias	0017	2000	250	0	0	0
Clutter native	0039	1100	138	0	1	−2
Clutter negative bias	0010	532	67	1	0	1

6.2 Full development outcomes

Table 2 preserves four full episodes on different tasks/seeds. Their native/bias differences do not estimate an adaptation effect. Jar native reaches the raw benchmark goal but fails the closed-lid sequence: requested unsafe unfinished, versus raw unsafe completion. Jar bias and Clutter native reach their full horizons, both valid outcomes rather than replaceable failures.

Development closeout contains ten starts, 4,394 controls, 4,464 environment steps and 26,987 known physics ticks. There are 551 physical-policy calls and 320 decoder calls. Encoding records 1,102 images, seven text calls, 23.519 s extraction and 38.443 s loading; physical-policy request wall time is 293.776 s. These totals precede the running comparison and are not final study costs. Callback-unobserved initialization physics remains unknown.

Clutter native has a 574.6 s cold startup versus its partner's 99.9 s. Logs identify a locked shared Kit cache and shader compilation. The episode then completes normally and retains its cost in capacity selection. This technical delay is distinct from its valid adverse physical outcome.

Table 3: Shared training initialization, two episodes per controller. SC/UC are safe/unsafe requested completion; SU/UU are safe/unsafe unfinished episodes. V counts violations and G episodes with a recorded grasp. Random here is the shared untrained conditional initializer, not a selected random-search finalist.

Controller	SC	UC	SU	UU	V	G
Native	0	0	1	1	1	1
Positive constant	0	0	1	1	1	0
Negative constant	0	0	2	0	0	0
Random conditional	0	0	0	2	2	0

6.3 Completed shared initialization

All eight shared initialization episodes reach the full 2,000-control horizon without requested or raw benchmark completion (Table 3). Each controller receives the same two nominal task/seed cases, costing 4,000 controls and 500 policy calls. Shared acquisition therefore costs 16,000 controls and 2,000 policy calls globally, attributed once within each optimizer's information budget.

The negative constant's utility sum zero leads native and positive at -2 and the random initializer at -4. It never grasps and contacts the target in only one of its two episodes. This lead concerns nonviolating unfinished execution, not improved safe completion. The frozen rule selects it as the fixed reference and initial incumbent of both searches. Native task 0019 goes off support while open at action 537; closure first becomes true at 638. Positive task 0009 instead loses uprightness at 1116 while still supported, with no recorded grasp. These are factual failure descriptions, not counterfactual explanations of a steering effect.

The initial robot input matches native in all six comparisons of a different controller on the same case. Both RGB views differ in every pair. Stored native-noise hashes and RNG transitions agree at all 1,500 paired query indices across those six streams. The common random stream is realized while observations remain different. These are correlated consistency checks over two cases, not independent outcome trials.

6.4 Exploratory gripper-coordinate diagnostic

Four generation-one incumbent rechecks subsequently evaluate the same negative constant on the new training seed panel. Each method scores $[0, -2]$: no requested or raw completion and one violation per pair. At this diagnostic cutoff, all twelve comparison episodes are unfinished, six violate, and only one ever grasps. Eight never command gripper closure; the other four each show actual finger closure. None of their arm proposals changes at the outer environment bounds. These observations do not identify a unique failure cause. In particular, actual offspring have not yet received a completed physical evaluation.

The frozen basis excludes direct support on the gripper-noise coordinate. During a diagnostic pause, a separately frozen 48-call follow-up tests coordinate seven (zero-based) on eight saved middle/last development frames, four per family from two episodes. Each input receives a native draw, an explicit same-noise repeat and offsets ± 0.5 and ± 2.0 on that coordinate at all sixteen chunk positions. No simulator or encoder runs. This raises total decoder requests to 368 while retaining the original audit and physical cohort unchanged. All eight additional same-noise identities pass at the original 10^{-5} tolerance.

Table 4: Supplemental gripper-noise influence. Entries count saved observations with at least one binarized gripper-command flip in the first eight decoded actions under either sign. The last column is the largest arm-target difference across those positions and the ± 2.0 probes. These are correlated decoder inputs, not physical outcome trials.

Family	Magnitude 0.5	Magnitude 2.0	Max arm (rad)
Jar	0/4	3/4	.05875
Clutter	0/4	4/4	.01773

The stronger offset changes gripper commands in both families (Table 4), with accompanying arm changes. Its positive sign opens the gripper in all four Clutter frames, including two factual held-target frames; its negative sign closes it in the two unfinished Jar-bias frames. Conditional timing and physical benefit remain untested. Three inputs already lie on violating factual trajectories, and no Jar input demonstrates a successful closure transition. A flip there cannot establish prospective violation prevention. This is influence in a candidate new representation, with no repair applied to the frozen physical controller.

An independent offline sensitivity analysis uses 3,334 factual Jar query rows and 205 Clutter rows from completed development/training episodes. The three persisted EA mutations change their parent codes by RMS .5410/.5825/.4130 on the Jar pool; only .990%/1.110%/.127% of their code components have magnitude at least .95. Inactive mutation or general tanh saturation is therefore unsupported as the current explanation. These are noise-space sensitivities on correlated factual inputs, with longer trajectories weighted more. They do not measure decoded actions, semantic competence or physical counterfactual benefit.

6.5 Search and independent comparison in progress

Optimizer and independent comparison results are pending. The completed records are preserved, and the original comparison resumed after the diagnostic under its unchanged protocol. Generation one is running on its predeclared seeds. No EA/random-search finalist is selected at this evidence cutoff. Training observations will not substitute for the protected sixteen-episode final check, and no optimizer advantage or transfer benefit is claimed.

7 LIMITATIONS AND REMAINING EVIDENCE

This interface can consume several family checkpoints, but the pilot fits only within Jar. A random four-direction basis may miss useful behaviour. A bounded noise offset need not imply safe contact dynamics. The 176-parameter domain is large relative to two-case candidate feedback. Noisy selection, adverse behaviour and identity finalists are valid outcomes. If a constant explains an improvement, conditioning is unestablished; if random search matches EA, evolutionary advantage is unestablished. An EA lead over uniform

random search combines parental reuse with a local Gaussian proposal distribution. Attributing it to evolutionary reuse additionally needs a fixed-parent Gaussian-search ablation and a matched local-search comparison under separate, prospectively funded evaluations.

An episodic parameter-REINFORCE interface over the same bounded matrix is implemented and CPU-checked, including interrupted-batch recovery. It has zero physical evaluations and is outside the frozen minimum allocation. It does not replace a competent state-conditioned DSRL-style latent SAC baseline. That baseline, matched tuning and independent optimizer replication are required before claiming practical evolutionary superiority. A direct-action residual baseline is additionally needed to attribute any advantage to the latent representation.

The cross-family target requires freezing one fitted controller and checking new tasks and an adaptation-held-out family with its own frozen base policy. Response-informed bases, persistence, sampling-distribution control and internal activation gates remain prospective, with no physical result here. A proposed 156-episode adaptation/transfer bridge is separately budgeted and unexecuted. Neither extensions nor tuning may consume the final checking allowance.

8 REPRODUCIBILITY AND PRESERVATION

The launch revision is `4783369c5b59`. Upstream pins are ManiGuard `6bf4c61a1f16`, BEHAVIOR `88454bd04f75` and openpi `215abfb217db` [5]. Separate uv-managed environments serve simulation (Python 3.10, Torch 2.6/CUDA 12.4), policy (Python 3.11, JAX .5.3) and CPU tooling. Comparative receipts retain actual GPU UUIDs, source hashes, causal features, decoded chunks, executed inputs and raw/requested outcome evidence. An independent reader verifies physical records before assigning fitness.

Code, compact evidence and frozen protocols are preserved in https://github.com/Sachin2911/Safety-Dial. The protocol is `configs/maniguard/latent-evolution/capacity-20261020-v2.json`; its study README documents reconstruction. Separate private Hugging Face model/dataset repositories store immutable stage archives with remote revision, privacy, size and SHA-256 verification. Full recovery reconstructs development records and all 32 identity pairs without new physics or model calls. Raw runs, licensed assets, credentials and base-policy weights are excluded from public Git.

9 CONCLUSION

Native-noise steering is implemented with conditional inputs and verified decoder influence in two family policies. Shared initialization yields no safe-completion gain; its highest-utility controller remains unfinished without grasping. The continuing frozen Jar comparison tests native execution, a tuned constant, evolution and matched random search while protecting independent checking and retaining unsuccessful candidates. Its result must determine the strength of the contribution, including a null or adverse finding. Ordinary learning, replication and transfer remain necessary for the broader claim.

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